SG No. 17 D_2^2 $P222_1$ [orthorhombic]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `2a`, site symmetry: `2..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-3]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,3]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`

* Wyckoff site: `2b`, site symmetry: `2..`

Table 4: Wyckoff bond: `2a@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3]`

Table 5: Wyckoff bond: `2b@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3]`

Table 6: Wyckoff bond: `4c@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`

4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[4]`

* Wyckoff site: `2c`, site symmetry: `.2.`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, y, \frac{1}{4}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[2,-4]`

Table 8: Wyckoff bond: `2b@2c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, y, \frac{1}{4}\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[0, -y, \frac{3}{4}\right]$	`[2,4]`

Table 9: Wyckoff bond: `4c@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, y, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, y, \frac{1}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[4]`

* Wyckoff site: `2d`, site symmetry: `.2.`

Table 10: Wyckoff bond: `2a@2d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[2,-4]`

Table 11: Wyckoff bond: `2b@2d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[2,4]`

Table 12: Wyckoff bond: `4c@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[3]`

continued ...

Table 12

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	[4]

* Wyckoff site: 4e, site symmetry: 1

Table 13: Wyckoff bond: 4a@4e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, -y, z + \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2} - z\right]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]